Multiobjective Spatiotemporal Subpixel Mapping for Remote Sensing Imagery

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***Abstract*— Subpixel mapping (SPM) aims to reconstruct a subpixel-level class distribution map from the pixel-level abundance maps, which is an under-determined problem that has nonunique solutions. To address this, the spatiotemporal SPM uses the abundance, spatial, and temporal constraints to reduce the uncertainty of the mapping solutions, so the spatiotemporal SPM is essentially a constrained optimization problem. However, it is hard to find the optimal weighting parameters to combine the three joint constraints. In addition, the existing spatiotemporal SPM methods mainly use the temporal information either for the unchanged subpixels detection or for the subpixel classification, which is insufficient in the utilization of the temporal information. In this article, a novel spatiotemporal SPM algorithm based on multiobjective optimization (STSPM_MO) is proposed. STSPM_MO is composed of an unchanged subpixels detection stage and a multiobjective spatiotemporal mapping stage. In the former stage, the historical thematic map is used for identifying the unchanged subpixels. In the latter stage, the historical thematic map is further used for providing the temporal dependence, so that the temporal information can be more fully utilized. Moreover, to solve the constrained optimization problem of the spatiotemporal SPM, the abundance, spatial, and temporal constraints are modeled as three objectives and are dynamically fused through the subfitness-based multiobjective evolution, to generate the optimal subpixel classification map. Both synthetic and real-data experiments have been conducted, and the results show the proposed method, and its two variants are superior, stable, and effective.**

***Index Terms*— Multiobjective optimization, remote sensing imagery, spatial dependence, subpixel mapping (SPM), temporal dependence.**

I. Introduction

SUBPIXEL mapping (SPM) is a technique that can predict the spatial distribution of land-cover classes at a subpixel scale for remote sensing imagery [1]. It was developed to alleviate the mixed pixel problem and improve the mapping resolution. Specifically, given a remote sensing image with a coarse spatial resolution, mixed pixels covering multiple classes are inevitable. The proportions (fractional abundances) of the component classes (endmembers) within each mixed pixel can be estimated by endmember extraction [2], [3] and spectral unmixing [4]. Based on the abundance maps, SPM is able to determine where the proportions of each class occur within the mixed pixel [5], [6]. In implementation, SPM divides the coarse pixel into a predefined number of subpixels, amounting to an increase in the spatial resolution, and then assigns an appropriate class label to each subpixel according to the abundance constraint. As a result, SPM can generate a classification map with a finer spatial resolution than the input image. However, the limited abundance information of a mixed pixel is not enough for the class label prediction of multiple subpixels because the abundance constraint of the mixed pixel can be satisfied by different class distributions of subpixels. Therefore, SPM is an under-determined problem, and its solution is not unique [7], [8], [9].

Adding auxiliary information as an a priori constraint is an effective way to make the underdetermined mapping problem well-posed. Actually, the development of SPM methods can be considered as a process of gradually increasing the auxiliary information. The concept of SPM proposed by Atkinson [5] is performed based on the spatial dependence assumption, so the land-cover map tends to be spatially smooth. Then, other auxiliary data have been introduced to provide additional spatial prior information for SPM, such as the multiple shifted images [7] and the spatial distribution dictionary [10]. SPM algorithms with auxiliary data are able to produce better mapping results with more details by learning useful spatial structure information. However, relying solely on the spatial dependence information tends to produce an over-smoothed class distribution map.

The temporal dependence between bitemporal images is another significant information source that can be utilized to reduce the uncertainty of SPM, i.e., if the images are temporally proximate, their land-cover distributions can be deemed to be similar. With the development of Earth observation systems, various remote sensing images with different spatial and temporal resolutions are now available, allowing the study of spatiotemporal SPM. Since the thematic map of a fine-resolution image acquired at a former time can be easily obtained from the Earth observation systems, this can provide plenty of temporal information, i.e., the fine spatial distribution details, for the SPM of the coarse image.

Many spatiotemporal SPM algorithms have been developed recently [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. Some of them use

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the temporal information at the change detection stage [13], [16], [24], [25]. That is, comparing the historical thematic map with the coarse-resolution abundance maps to mark the unchanged subpixels, then changed subpixels are labeled by the traditional mapping method. Other spatiotemporal SPM algorithms use the temporal information at the mapping stage [12], [15], [17], [18], [18], [20], [21], [22], [23], i.e., using the historical thematic map to characterize the temporal constraint of the mapping model. Specifically, these approaches construct the mapping model with three constraints: spectral, spatial, and temporal, and then combine the three constraints into a cost function by linear weighting. However, there are some contradictions between the three constraints. For example, optimizing the abundance term tends to recover the details of land-cover distribution in the current time phase, optimizing the spatial term tends to generate a uniform land-cover distribution, and optimizing the temporal term tends to retain the details of land-cover distribution from historical time phases. Therefore, the multiple constraints often cannot achieve optimality at the same time. In addition, the weights are sensitive to different inputs, and they need to be empirically preset. Fine tuning the weight parameters significantly affects the fusion performance of the abundance, spatial, and temporal constraints, which, in turn, has a notable impact on the mapping accuracy [15]. Therefore, the mapping accuracy is highly sensitive to the empirical weights. In addition, from the above survey, we can see that the temporal information is only used at one stage in the presented spatiotemporal SPM methods, i.e., either at the change detection stage or at the mapping stage. Therefore, the utilization of the temporal information in these methods is insufficient.

To further improve the mapping accuracy, it is necessary to construct a model to handle the contradict constraints of spatiotemporal SPM and to find an optimization method without sensitive weight to integrate and balance the abundance, spatial, and temporal information. Since the constraint satisfaction of SPM is similar to the multiobjective optimization, with both involving the simultaneous optimization of a number of contradictory functions [27], [28], multiobjective optimization has been successfully applied to solve the constrained optimization problem of single-image SPM [multiobjective optimization based subpixel mapping algorithm (MOSM) and multiobjective optimization based spectral unmixing and subpixel mapping algorithm (MO_SUSM)] [29], [30]. In MOSM [29], the spectral and spatial constraints are treated as two objective functions and are optimized simultaneously by the multiobjective evolutionary algorithm (MOEA) to avoid the sensitive weights. In MO_SUSM [30], the spectral unmixing model is integrated into the spectral constraint of SPM to reduce the cumulative errors of unmixing and SPM. Both MOSM and MO_SUSM prove that the information of the SPM constraints can be automatically integrated to search for the optimal subpixel distribution. However, the temporal dependence is not taken into account in MOSM and MO_SUSM, resulting in a relatively poor mapping accuracy.

In this article, a novel spatiotemporal SPM algorithm based on multiobjective optimization (STSPM_MO) is proposed. The proposed method is composed of two stages: 1) the unchanged subpixels detection and 2) the multiobjective spatiotemporal mapping. In the former stage, a subpixel change detection strategy [13] is adopted to assign the class labels to the unchanged subpixels by comparing the coarse abundance maps with the historical thematic map. In the latter stage, the constrained optimization problem of spatiotemporal SPM is transformed into a multiobjective optimization problem (MOP) by modeling the abundance, spatial, and temporal constraints as three objectives, and the subfitness-based MOEA [29] is applied to integrate the multiple objectives and generate the class distribution with different trade offs between the three objectives. The contributions of this work are summarized as follows.

1) *Full Exploitation of the Temporal Information:* The temporal information, i.e., the fine spatial distribution of the historical thematic map, is utilized in the two stages of the proposed method. In the maps fusion stage, the thematic map is used to identify the unchanged subpixels and provides the class labels for them. In the multiobjective optimization stage, the thematic map is used to characterize the temporal prior term of the multiobjective mapping model, to constrain the underdetermined SPM problem. Thus, the valid information of the thematic map can be further exploited for the allocation of other subpixels.
2) *Multiobjective Spatiotemporal SPM Algorithm:* Multiobjective optimization is applied to solve the constrained problem of spatiotemporal SPM. A multiobjective mapping model with three objectives—data fidelity term, spatial prior term, and temporal prior term—is proposed, representing the abundance constraint, spatial dependence, and temporal dependence, respectively. The multiple contradictory objectives are simultaneously optimized by the subfitness-based MOEA to search for the class distribution map. During the optimization, a refinement strategy is designed to incorporate the unchanged distributions obtained at the map fusion stage, and the evolutionary operators, including clone, crossover, and mutation are conducted at each iteration, thereby dynamically integrating the information of the three objectives.

The rest of this article is organized as follows. The basic concepts related to spatiotemporal SPM are briefly introduced in Section II. The framework and details of the STSPM_MO method are described in Section III. The experiments undertaken to verify the performance of the proposed algorithm are described in Section IV, followed by the parameter analysis and the computation time. A discussion is also provided in this section. Conclusion is drawn in Section V.

II. Related Works

A. Subpixel Mapping Problem

Based on the abundance maps obtained by spectral unmixing, SPM aims to generate a classification map at the subpixel scale. An intuitive example is shown in Fig. 1. Suppose that y^c is the abundance map of class c with a size of $[m, n]$, under the mapping scale S, each coarse pixel of the abundance map is then separated into S^2 subpixels, which make up a mapping

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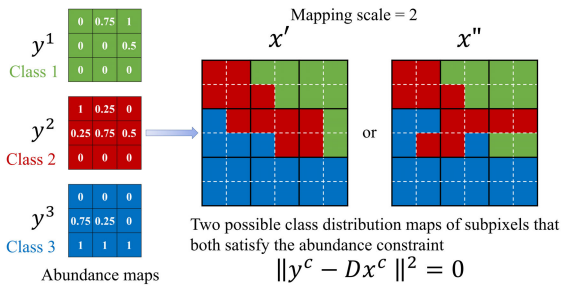


Fig. 1. Illustration of the SPM problem. The abundance constraint can be satisfied by different solutions.

block. x^c is the subpixel distribution map of class c with a size of $[M, N]$, where $M = m \times S$ and $N = n \times S$. The resultant classification map is denoted by $x = \{x^c\}, c = 1, 2, \ldots, C$, where C is the number of component classes. The abundance constraint used to describe the relationship between y^c and x^c can be modeled by (1). D is a downsampling matrix with a size of $[m \times n, M \times N]$. $x^c(i)$ denotes the ith subpixel in x^c, and $i = 1, 2, \ldots, M \times N$

$$y^c = Dx^c, \quad c = 1, 2, \ldots, C$$
$$x^c(i) = \begin{cases} 1, & \text{if subpixel } x(i) \text{ is allocated to class } c \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

To better resist the unmixing error of the abundance maps, the abundance constraint is relaxed to [31, eq. (2)] and [32]. x^c is the distribution map, in which $x^c(i)$ represents the probability of class c distributed in the ith subpixel of x^c. The mapping solution is supposed to minimize $R(x^c)$

$$R\left(x^c\right) = \left\|y^c - Dx^c\right\|^2, \quad 0 \le x^c(i) \le 1 \quad i = 1, 2, \ldots, M \times N. \tag{2}$$

From Fig. 1, we can see that two possible class distribution maps x' and x'' both satisfy the abundance constraint. Without other knowledge, it is difficult to decide which of them is the best. To regularize the undetermined SPM problem, the spatial constraint is needed.

B. Spatial Dependence

The spatial dependence assumption is the foundation of SPM and is used as a priori knowledge to obtain a unique mapping solution. Spatial dependence means that class labels of the adjacent subpixels are likely to be the same. Mathematical models have been proposed to quantify spatial dependence, such as the spatial attraction model (SASM) [15], [17], [18], [19], [33] and the markov random field (MRF) model [34], [35]. In [20], [29], and [31], the Laplacian prior defined by (3) is adopted as the spatial constraint $U(x^c)$, which uses a 2-D convolution kernel Q to measure the accumulated abundance gradient within each subpixel block. Under the abundance constraint, the unique solution of SPM can be obtained by minimizing $U(x^c)$

$$U\left(x^c\right) = \left\|Qx^c\right\|^2. \tag{3}$$

In the case of Fig. 1, the class distribution map x' would be better than x'', according to the spatial constraint. However, the spatial dependence ignores the small objects and narrow objects. The solution that maximizes the spatial dependence fails to reconstruct the complicated distributions. For example, if the complex class distribution x'' in Fig. 1 is the correct one, then without additional information, the mapping problem cannot be solved based on the spatial dependence.

C. Temporal Dependence

Temporal dependence is another priori knowledge derived from a historical thematic map with finer spatial resolution. Temporal dependence means that the class distribution is more likely to remain unchanged over a short period of time. Therefore, in the SPM process, there can be many effective land-cover details worth learning from the historical thematic map. Some spatiotemporal SPM methods [15], [17], [19] characterize the temporal dependence and spatial dependence in the same way, i.e., modeling them as the attractiveness. Other methods [18], [20] determine the temporal dependence $T(x^c)$ by minimizing the difference between the class distribution map x^c and the historical thematic map x_h^c. The mapping solution is supposed to minimize $T(x^c)$

$$T\left(x^c\right) = \left\|x^c - x_h^c\right\|^2$$
$$x_h^c(i) = \begin{cases} 1, & \text{if subpixel } x_h(i) \text{ is occupied by class } c \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Temporal dependence can help us to reconstruct complex class distribution details, and combining it with the spatial dependence and abundance constraint can obtain a unique and accurate mapping solution. For example, given the historical class distribution x_h in Fig. 2, the mapping solution x'' with higher temporal dependence is considered to be better than the solution x' in Fig. 1.

D. Spatiotemporal Subpixel Mapping

The flowchart of spatiotemporal SPM is depicted in Fig. 2. With the input of the abundance maps of a coarse image acquired at time $T2$, and the historical thematic map of a finer-resolution image acquired at a former time $T1$, the goal of spatiotemporal SPM is to produce a finer classification map for the $T2$ phase under the abundance, spatial, and temporal constraint.

Some works have been done to model and utilize the three constraints in different ways. For example, the spatiotemporal SPM algorithm based on swarm intelligence theory [spatiotemporal subpixel mapping based on clonal selection (SSM_CS)] proposed in [13] integrates the temporal information into the coarse abundance maps before SPM; then, the mapping result is obtained by the clonal selection algorithm. Xu and Huang [12] proposed a spatiotemporal SPM method based on the conventional pixel swapping algorithm (STPSA), in which the abundance and temporal constraints are calculated as the attractiveness before swapping the pair of subpixels. Ling et al. [19] extended the spatial dependence used in the pixel swapping algorithm to a spatiotemporal dependence optimized by a simulated annealing algorithm [superresolution mapping based on spatiotemporal dependence (SRM_STD)]. In [15], the temporal dependence index based on the attraction is proposed, and its influence was adjusted based on

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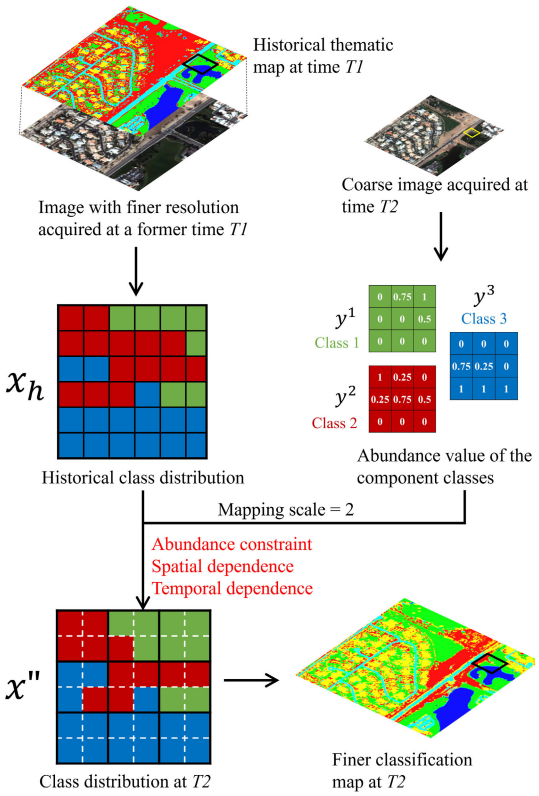


Fig. 2. Flowchart of spatiotemporal SPM. The abundance, spatial, and temporal constraints are fused to produce an accurate mapping solution.

the time interval and changed ratio between the bitemporal images [spatiotemporal subpixel geographical evolution mapping (STGEM)]. He et al. [20] integrated the abundance, spatial, and temporal information by a maximum a posteriori (MAP) model (SST_MAP), and a temporal regularization term is added into the MAP-based SPM model. For the application, the subpixel land-cover change mapping method proposed in [16] fused the former fine-resolution thematic map and the latter coarse-resolution abundance maps to predict the subpixel changes (SLCCMs). Li et al. [18] proposed a spatiotemporal super-resolution mapping algorithm (STSRM), which constructed the temporal constraint using a few fine spatial resolution land-cover maps that pre- and post-date the series of coarse spatial resolution images. Wang et al. [17] designed a spatiotemporal SPM method for time-series images (SPT), using the subpixel-wise correlation to construct the temporal dependence, and the spatial and temporal constraints were combined into an energy function optimized by simulated annealing algorithm.

The abovementioned spatiotemporal SPM algorithms have proved the validity and significance of the temporal information. However, utilization of the temporal information in these algorithms is insufficient, because the historical thematic map is only used at one stage in SPM, i.e., either for change detection with the abundance maps or for modeling the temporal dependence. In addition, the previous spatiotemporal SPM methods [15], [17], [19], [20] generally combine the three constraints ($R(x^c)$, $U(x^c)$, and $T(x^c)$) into a cost function $F(x^c)$

$$\min \; F\left(x^c\right) = R\left(x^c\right) + \alpha U\left(x^c\right) + \beta T\left(x^c\right). \tag{5}$$

However, the two weights (α and β) are sensitive to different inputs, so the optimal weights are difficult to establish. In addition, the preset weights fail to dynamically balance the three constraints in the mapping process. Furthermore, the presented methods ignore the intrinsic contradictory of these constraints. Therefore, it is necessary to find a better way to incorporate their information and balance their influence.

III. Multiobjective Spatiotemporal Subpixel Mapping

To better solve the constrained problem in spatiotemporal SPM, a novel STSPM_MO is proposed (Fig. 3). We can see that a multiobjective mapping model is constructed to present the three constraints of spatiotemporal SPM. An unchanged subpixel detection stage is first conducted to preserve the unchanged spatial distribution. At the latter mapping stage, the MOEA is applied to balance the three constraints of spatiotemporal SPM and produce the subpixel spatial distribution maps by optimizing the multiobjective model. The final classification map is determined by integrating the subpixel spatial distribution maps of all classes. In particular, the temporal information of the historical thematic map is both utilized in the unchanged subpixels detection stage and the mapping stage of STSPM_MO. Pseudocode of STSPM_MO are provided in **Algorithm 1**, and details are described as follows.

A. Multiobjective Model for the Spatiotemporal Subpixel Mapping

To construct a suitable model to describe the constrained spatiotemporal SPM problem, we have to consider the contradiction between the abundance, spatial, and temporal constraints. Specifically, the abundance constraint tends to maintain the distribution details of the current phase. The spatial constraint tends to generate a smooth class distribution map. The temporal constraint tends to preserve the historical land-cover details. Therefore, for a mixed pixel, the three constraints usually cannot achieve optimality at the same time.

Since the constraint satisfaction of the spatiotemporal SPM problem and the multiobjective optimization are very much alike, it is natural to model it as an MOP. MOP refers to a problem involving more than one contradictory objective functions that need to be minimized or maximized [36]. Its general form is

$$\min F(z) = \min \left(f_1(z), f_2(z), \ldots, f_k(z)\right)^{\mathrm{T}} \tag{6}$$

where $z = (z_1, \ldots, z_p)^{\mathrm{T}} \in \Omega$ is the solution vector. Multiobjective optimization model $F(\cdot)$ has multiple objective functions $f_1(\cdot), f_2(\cdot), \ldots, f_k(\cdot)$ that need to be simultaneously minimized. With the solution z, $F(z) \in \Re^k$ is a vector of k objective function values. Since there is no solution that can make the multiple objectives simultaneously reach optimality, the optimal solution for the MOP consists of a number of solutions with different trade offs between each objective.

As to the spatiotemporal SPM problem, its abundance, spatial, and temporal constraints are converted into three objective functions in the proposed multiobjective model (7),

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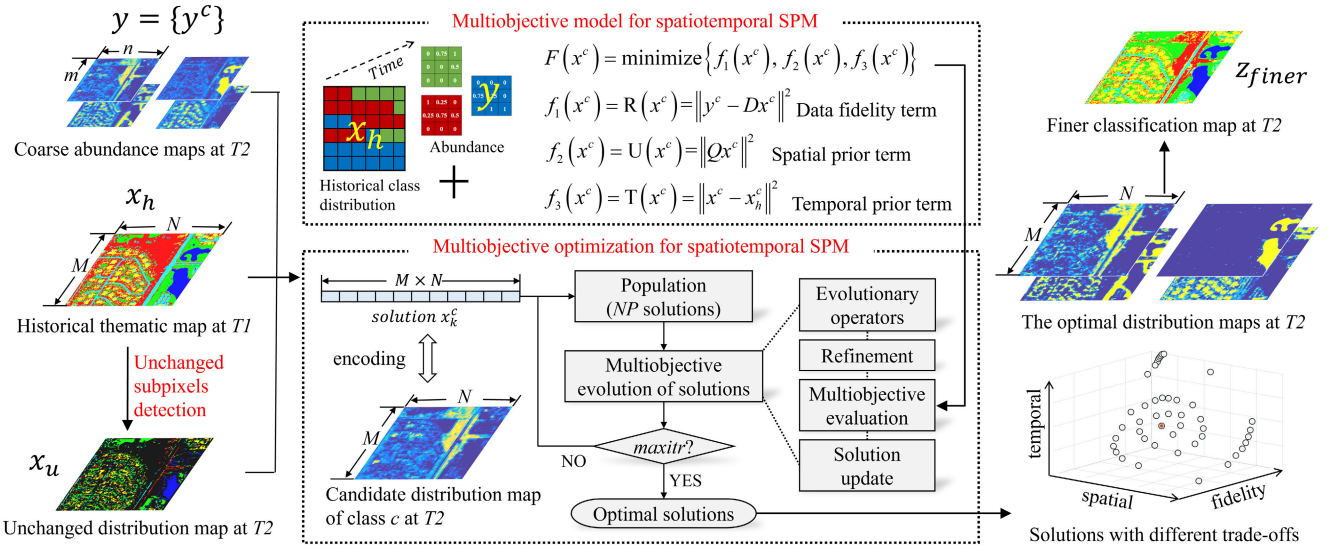


Fig. 3. Framework of STSPM_MO. With the input of the abundance maps and historical thematic map, an unchanged subpixel detection step is conducted to maintain the unchanged distributions. At the mapping step, a multiobjective mapping model is proposed, and then, the MOEA is applied to optimize the model and produce the trade-off distribution maps. The final classification map is obtained by integrating the distribution maps.

where x^c represents the distribution map of class c. y^c denotes the abundance map of class c. D is the downsampling matrix

$$\begin{aligned} \min\ F(x^c) &= \min\left(f_1(x^c), f_2(x^c), f_3(x^c)\right)^{\mathrm{T}} \\ \text{data fidelity term } f_1(x^c) &= R(x^c) = \left\|y^c - Dx^c\right\|^2 \\ \text{spatial prior term } f_2(x^c) &= U(x^c) = \left\|Qx^c\right\|^2 \\ \text{temporal prior term } f_3(x^c) &= T(x^c) = \left\|x^c - x_h^c\right\|^2. \end{aligned} \tag{7}$$

The data fidelity term is defined based on the abundance constraint (2) to maintain the spatial proportion of each class within the coarse pixel. The spatial prior term [37] is adopted to incorporate the spatial information of neighbor subpixels. The temporal prior term [20] is used to incorporate the distribution information of the historical thematic map.

B. Unchanged Subpixels Detection

The unchanged subpixels detection strategy [16], [25], as illustrated in Fig. 4, is designed to produce the unchanged distribution map $x_u = \{x_u^c\}$ by comparing the coarse-resolution abundance maps $y = \{y^c\}$ with the finer-resolution thematic map $x_h = \{x_h^c\}$. To be more specific, the historical thematic map x_h is first down sampled to generate the auxiliary abundance maps $y_h = \{y_h^c\}$ for each class c. The difference maps $y_d = \{y_d^c\}$ are then obtained by subtracting y_h from y. The unchanged distribution map x_u is determined by integrating x_h with y_d using (8), where $y_d^c(i)$ is the ith pixel in y_d^c. $x_h^c\{i\}$ and $x_h^c\{i\}$ are the corresponding ith mapping blocks of x_h^c and x_u^c, respectively,

$$x_u^c\{i\} = \begin{cases} x_h^c\{i\}, & \text{if } y_d^c(i) \geq 0 \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

With the high-quality abundance maps, the unchanged land-cover details can be preserved by the unchanged subpixel detection. However, affected by the unmixing error of the abundance maps, many unchanged subpixels can be misjudged as changed. Therefore, the distribution information of the historical thematic map is still worth learning in the multiobjective mapping stage.

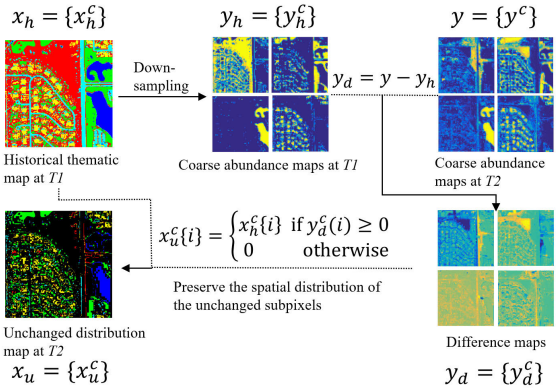


Fig. 4. Unchanged subpixels detection. The abundance maps are compared with the historical thematic map, to identify the unchanged spatial distributions.

C. Multiobjective Optimization for Spatiotemporal Subpixel Mapping

At this mapping stage, the subpixel spatial distribution of the possible changed land cover is determined by the MOEAs. MOEAs are a sort of optimization algorithm designed for solving MOPs [38], which can take various information (e.g., abundance, spatial, and temporal information) into account by simultaneously optimizing the multiple objective functions, regardless of the sensitive weights. In this article, the subfitness-based MOEA proposed in [29] is employed to solve the MOP of spatiotemporal SPM. In addition, a cloning operator and a refinement operator are added to the subfitness-based MOEA for better performance.

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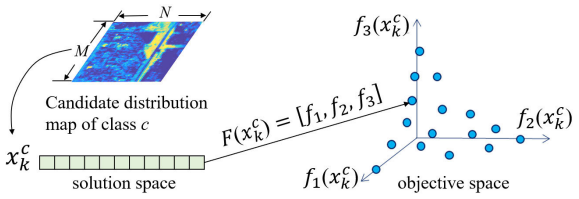


Fig. 5. Population initialization with different trade-off mapping solutions. Each solution can be reshaped to a distribution map.

With the input of the abundance maps $y = \{y^c\}$ of the coarse remote sensing image, a finer-resolution historical thematic map $x_h = \{x_h^c\}$, and the unchanged distribution map $x_u = \{x_u^c\}$, the optimization process is performed for each abundance map to generate the subpixel-scale distribution map for each class.

1) Population: The population comprises NP solutions, i.e., POP $= \{x_k^c\}$, $k = 1, 2, \ldots, NP$. Each solution is an $M \times N$ vector that can be reshaped to a candidate distribution map of class c, and it is randomly initialized around the distribution map obtained by nearest neighbor interpolation of abundance map y^c. As depicted in Fig. 5, based on the multiobjective spatiotemporal SPM model (7), a candidate distribution map in the solution space can be mapped to a 3-D point in the objective space. After population initialization, NP points are randomly distributed in the objective space, representing different trade offs between the abundance, spatial, and temporal constraints.

2) Evolutionary Operators: The evolutionary process is conducted on each solution x_k^c in POP to optimize the number and position of the subpixels that are occupied by class c. A flowchart of the MOEA is shown in Fig. 6. For a better exploitation of the solution space, the solution x_k^c is first cloned to generate t copies. Each copy then interacts with two distinct solutions x_{p1}^c, x_{p2}^c using the crossover and mutation operators in (9). After this, t child solutions x_G^c can be produced

$$\begin{aligned} x_{\text{mu}}^c &= x_k^c + F \times \left(x_{p1}^c - x_{p2}^c\right) \\ x_{\text{de}}^c(j) &= \begin{cases} x_{\text{mu}}^c(j), & \text{if rand} \le \text{CR or } j = j_{\text{rand}} \\ x_k^c(j), & \text{otherwise.} \end{cases} \\ x_G^c &= \begin{cases} N\left(x_{\text{de}}^c(j),\ \delta^2\right), & \text{if rand} < p \\ x_{\text{de}}^c, & \text{otherwise} \end{cases} \end{aligned} \tag{9}$$

where t is the number of clone solutions. $p1, p2 \in [1, NP]$ are two random integers. $x_k^c(j)$ is the jth subpixel in the solution x_k^c. The scale factor $F \in [0, 1]$ and crossover rate CR $\in [0, 1]$ are the evolutionary control parameters. $j_{\text{rand}} \in [1, NP]$ is a random integer, and $N(a, \delta^2)$ is a normal distribution with a mean of a and a standard deviation of δ. $p = 0.1$ denotes the mutation probability.

3) Refinement: In order to maintain the labels of the unchanged subpixels, each child solution is further refined by the unchanged distribution map

$$x_R^c(j) = \begin{cases} 1, & \text{if } x_u^c(j) = 1 \\ x_G^c, & \text{otherwise.} \end{cases} \tag{10}$$

Algorithm 1 Pseudocode of STSPM_MO

Require: $y = \{y^c\}$, $x_h = \{x_h^c\}$, $c = 1, 2, \ldots, C$
Ensure: $x = \{x^c\}$, z_{finer}
Para initialization S, D, $NP = 50$, λ, $F = 0.8$, $CR = 0.5$, $t = 5$, $itr = 1$, $maxitr = 30$, λ
Step 1. Unchanged subpixels detection.
y_h = down-sampled(x_h), y_d = y-y_h.
Produce the unchanged distribution map $x_u = \{x_u^c\}$ by (6).
for each class c **do**
Step 2. Population.
x_0^c = nearest neighbor interpolation(y^c).
Generate an initial population randomly $POP = \{x_k^c\}$, $k = 1, 2, \ldots, NP$.
$x_k^c = \min(x_0^c) + rand \times (\max(x_0^c) - \min(x_0^c))$
x_k^c is assigned to the search direction λ_k.
Step 3. Multiobjective evaluation.
for i $= 1, 2, \ldots, NP$ **do**
Divide solution into mapping blocks.
$x_k^c = \{x_k^c[i]\}, i = 1, 2, \ldots, m \times n$.
Evaluate mapping blocks by (7).
Calculate the subfitness $F'(x_k^c[i])$ by (11).
end for
repeat
Step 4. Evolutionary operators.
for each solution x_k^c, $k = 1, 2, \ldots, NP$ **do**
x_k^c is cloned to generate t copies.
for each copy of x_k^c **do**
Two distinct solutions x_{p1}^c, x_{p2}^c are randomly selected from POP.
Produce child solution x_G^c by (9).
Step 5. Refinement.
Produce refined solution x_R^c by (10).
x_R^c is corrected to the range of [0, 1].
Step 6. Multiobjective evaluation.
x_R^c is assigned to the search direction λ_k.
Divide solution into mapping blocks.
$x_R^c = \{x_R^c[i]\}, i = 1, 2, \ldots, m \times n$.
for each mapping block $x_R^c[i]$ **do**
Evaluate mapping block by (8).
Calculate the subfitness $F'(x_R^c[i])$ by (11).
Step 7. Solution update.
if $F'(x_R^c[i]) < F'(x_k^c[i])$ **then**
$x_k^c[i] \leftarrow x_R^c[i]$
end if
end for
end for
end for
$itr = itr + 1$;
until $itr = maxitr$
Select the best distribution map x_{best}^c from the final POP by visual judgment.
end for
Generate subpixel classification map z_{finer} by integrating the best distribution maps of C classes.
$z_{finer} = \text{argmax}\{x_{best}^c\}$, $c = 1, 2, \ldots, C$.

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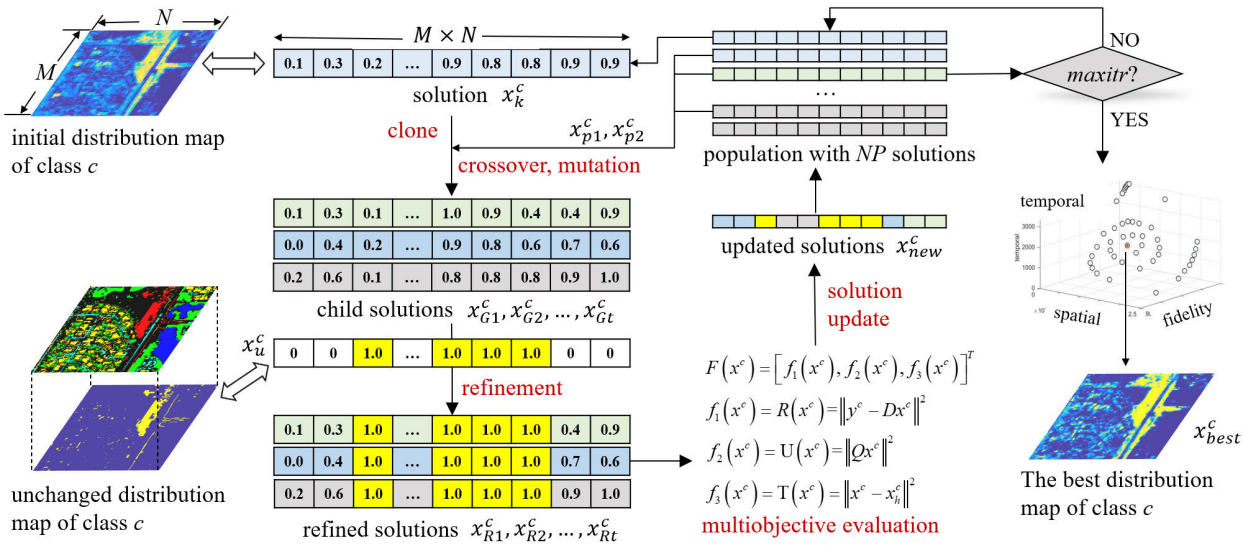


Fig. 6. MOEA for the optimization of the spatiotemporal SPM model. The trade-off mapping solutions interact with each other during the evolution, which enables dynamic fusion of the abundance, spatial, and temporal information.

4) Multiobjective Evaluation: To evaluate the quality of the t refined solutions, each high-dimensional solution x_R^c is separated into $m \times n$ mapping blocks $x_R^c[i], i = 1, 2, \ldots, m \times n$. $[m, n]$ is the image size of y^c. Subfitness of each mapping block $x_R^c[i]$ is evaluated by the cost function (11) and the multiobjective mapping model (7) based on the Tchebycheff approach. The mapping block with a lower cost function value has a higher subfitness, indicating a better quality

$$F'\left(x_R^c[i]\right) = g^{\text{te}}\left(x_R^c[i] \mid \lambda^R, r^*\{i\}\right) = \max_{k=1,2,3}\left\{\lambda_k^R \left| f_k\left(x_R^c[i]\right) - r_k^*\{i\}\right|\right\} \quad (11)$$

where $f_k(\cdot)$ is the kth objective term in (7). $\lambda = \{\lambda^1, \lambda^2, \ldots, \lambda^{NP}\}$ is a set of 3-D vectors that are uniformly initialized in the objective space. λ^R defines the search vector of the solution x_R^c while satisfying the constraint $\sum_{k=1}^3 \lambda_k^R = 1$. $r^*\{i\}$ is the reference point of the ith mapping block, which is defined by the minimum value of the multiobjective model (7) of the ith mapping block found so far.

5) Solution Update: After evaluation, the t refined solutions x_R^c are then used to update the old solution x_k^c by comparing their subfitness of the corresponding mapping blocks. The mapping block with a higher subfitness will be preserved to the new solution x_{new}^c. We refer the reader to [19] for more details on this update strategy, which can ensure the reproduction of a new solution with a higher quality than the older one.

After updating the whole population, NP new solutions can be generated. From Fig. 6, we can see that different mapping solutions interact with each other during the evolution. Since these solutions represent different trade offs between the multiple objectives, their interaction enables dynamic fusion of the abundance, spatial, and temporal constraints.

The evolutionary process is repeated until the stopping condition is satisfied, i.e., the number of iterations reaches the maximum, in this article maxitr = 30. The final population comprises NP optimal solutions, i.e., the distribution maps of class c. These solutions are incomparable, because none of them can outperform the other solutions on all three objectives. Since the distribution maps provide the visual results for different balances of the multiple constraints, the best distribution map x_{best}^c can be selected from the final population by simple visual judgment.

The optimization process is terminated when the distribution maps of all the classes have been predicted. The final subpixel classification map z_{finer} is determined by integrating the best distribution maps. The jth subpixel $z_{\text{finer}}(j)$ is allocated to the class with the highest distribution probability, i.e., $z_{\text{finer}}(j) = \text{argmax}\{x_{\text{best}}^c(j)\}$, $c = 1, 2, \ldots, C$.

IV. Experiments and Analysis

To assess the performance of the proposed STSPM_MO approach, the experiment based on the synthetic abundance maps simulated from Landsat-8 images was undertaken, as well as two real-data experiments based on Landsat-8, Gaofen-2, and Sentinel-2A images. The results are quantitatively evaluated by the producer's accuracy (PA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient. Six SPM algorithms related to our work were selected as comparative methods: MAP model [31], MOSM [29], SLCCM [16], SRM_STD [19], SSM_CS [13], SPT [17], STSRM [18], and SST_MAP [20]. The comparative methods are briefly described as follows.

MAP: The MAP model combines the abundance constraint (2) with the Laplacian-based spatial constraint (3) to regularize the SPM problem.

MOSM: The MOEA-based SPM method simultaneously optimizes the abundance constraint (2) and the Laplacian-based spatial constraint (3).

SLCCM: The spatiotemporal SPM method for subpixel land-cover change mapping, which observes the abundance constraint (1) and determines the subpixel labels by the subpixel change detection.

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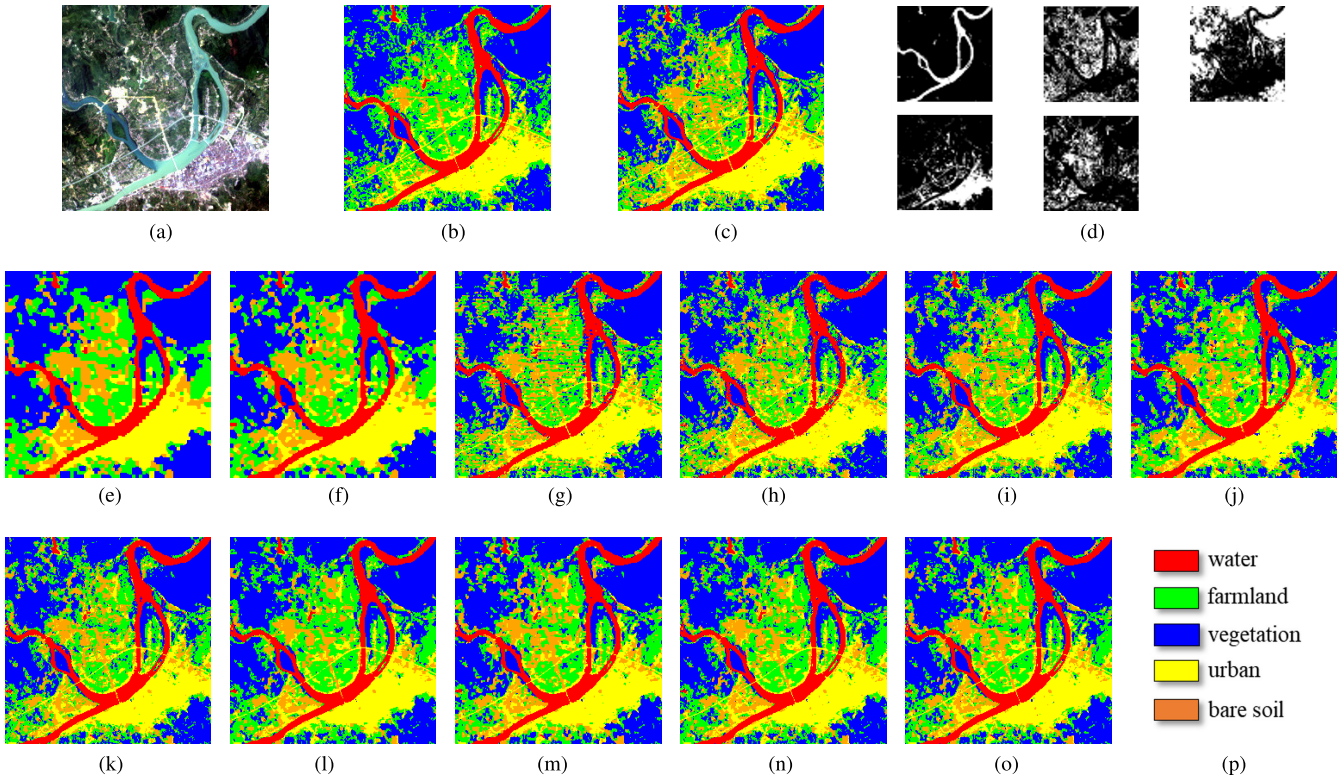


Fig. 7. SPM results for the synthetic Cili dataset. (a) July 13, 2013, Landsat-8 image. (b) July 13, 2013, historical thematic map. (c) August 7, 2013, reference map. (d) August 7, 2013, abundance maps were obtained by downsampling the reference map. (e) MAP. (f) MOSM. (g) SLCCM. (h) SRM_STD. (i) SSM_CS. (j) SPT. (k) STSRM. (l) SST_MAP. (m) SSM_MO. (n) SST_MO. (o) STSPM_MO. (p) Color legend of five land-cover types.

SRM_STD: Spatiotemporal SPM method extends the spatial dependence of the classical pixel swapping algorithm to a spatiotemporal dependence by integrating the historical thematic map, optimized by simulated annealing.

SSM_CS: An extension of SLCCM, applying the subpixel change detection strategy to incorporate the temporal information, then maximizing the spatial attraction by the clonal selection algorithm.

SPT: Spatiotemporal SPM method for time-series images, which uses the subpixel-wise correlation to construct the temporal constraint as distance-weighted attractiveness, optimized by the simulated annealing.

STSRM: The STSRM incorporates the pixel spectral and land-cover spatial and temporal information into an objective function and is optimized by the iterative conditional mode.

SST_MAP: The latest spatiotemporal SPM algorithm that uses the MAP model to combine the abundance constraint (2) with the spatial constraint (3) and the temporal constraint (4).

In order to verify the role of the temporal information at each stage of STSPM_MO, two variant algorithms (SSM_MO and SST_MO) were also tested in the experiments. The former is a variant that incorporates the temporal information only in the unchanged subpixels detection stage, while the latter is another uses the temporal dependence only in the multiobjective mapping stage. Moreover, the superiority of the MOEA in solving the constrained problem of spatiotemporal SPM can be proved by comparing SSM_CS with SSM_MO or SST_MAP with SST_MO.

In addition, a sensitivity analysis of the control parameters was conducted to demonstrate the stability of the STSPM_MO approach. The comparison of the computation time and a further discussion are also provided.

A. Experiments on Synthetic Dataset

In the synthetic experiment, the input abundance maps were simulated by downsampling the fine-resolution land-cover map. To be more specific, a multispectral image acquired by the Landsat-8 satellite, covering Cili County, was adopted as the original fine-resolution image. The acquisition time was August 7, 2013. In the image subset of 240 × 240 × 7 pixels, there were five land-cover types: water, farmland, vegetation, urban, and bare soil. The support vector machine (SVM) classification algorithm was applied to produce the reference map [Fig. 7(c)] for SPM, whose classification accuracy is 93.91%. Under the mapping scale of 4, the reference map was downsampled to simulate the synthetic abundance maps of 60 × 60 × 5 [Fig. 7(d)]. For the spatiotemporal SPM, another image of Landsat-8 acquired on July 13, 2013, was used as the historical image [Fig. 7(a)], whose SVM classification map was adopted as the historical thematic map [Fig. 7(b)]. The bitemporal images of Landsat-8 were matched at a subpixel level by the environment for visualizing images (ENVI). The change ratio between the multitemporal land-cover maps [Fig. 7(b) and (c)] was 20.54%. For water, farmland, vegetation, urban, and bare soil, the respective change ratios are 4.94%, 40.90%, 13.89%, 9.35%, and 15.30%. The SPM

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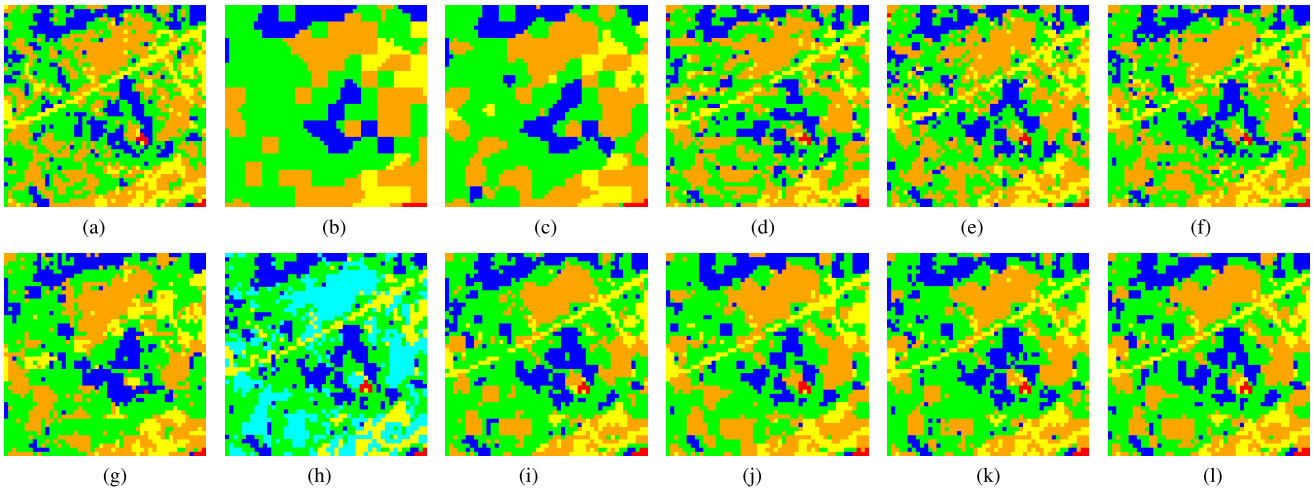


Fig. 8. Zoomed-in areas of the SPM results for the synthetic Cili dataset. (a) Reference. (b) MAP. (c) MOSM. (d) SLCCM. (e) SRM_STD. (f) SSM_CS. (g) SPT. (h) STSRM. (i) SST_MAP. (j) SSM_MO. (k) SST_MO. (l) STSPM_MO.

TABLE I
MAPPING ACCURACIES FOR THE SYNTHETIC CILI DATASET

	MAP, 2015	MOSM, 2018	SLCCM, 2010	SRM_STD, 2014	SSM_CS, 2016	SPT, 2016	STSRM, 2017	SST_MAP, 2019	**SSM_MO**	**SST_MO**	**STSPM_MO**
Water	95.32	96.17	96.86	94.27	97.34	96.12	98.41	98.96	98.22	99.14	99.14
Farmland	62.81	63.55	69.47	67.18	68.41	68.10	72.53	79.10	72.19	76.21	74.34
Vegetation	87.26	87.60	87.64	89.18	89.99	88.17	91.36	91.80	91.68	92.25	92.91
Urban	76.47	78.97	84.17	86.96	88.21	83.34	89.79	89.89	88.86	91.03	90.95
Bare soil	57.33	60.89	63.46	63.13	65.75	57.37	62.35	53.30	68.51	61.60	65.98
OA (%)	76.01	77.34	79.98	80.12	81.61	78.68	82.84	83.12	83.71	84.14	**84.65**
AA (%)	75.84	77.44	80.32	80.14	81.94	78.62	82.89	82.61	83.90	84.05	**84.67**
Kappa	0.6839	0.7018	0.7375	0.7388	0.7584	0.7192	0.7741	0.7770	0.7858	0.7910	**0.7976**

methods attempted to recover the fine-resolution classification map based on the coarse-resolution abundance maps.

The synthetic dataset is used to evaluate the performance of the SPM algorithms under undisturbed conditions free from the unmixing errors. As can be seen from maps and zoomed-in areas in Figs. 7 and 8, the traditional SPM methods (MAP and MOSM) produce over-smooth results and lose most of the details in urban, farmland, and vegetation classes. This is because the input abundances are accurate, but MAP and MOSM rely on the spatial prior term to judge the number and position of each class. Compared with the traditional SPM methods, the spatiotemporal SPM methods (SLCCM, SRM_STD, SSM_CS, SPT, STSRM, SST_MAP, SSM_MO, SST_MO, and STSPM_MO) benefit from the temporal information and can better reconstruct the spatial distribution. Based on the clean abundance input, all the spatiotemporal SPM methods produce satisfactory results with little visual difference. As the zoomed-in areas shown in Fig. 8, the proposed methods (STSPM_MO and the two variants of SSM_MO and SST_MO) generate more accurate maps with finer boundaries, especially for the urban, bare soil, and water classes with small change ratios. This is because the MOEA can dynamically fuse the abundance, spatial, and temporal constraints during the optimization of SPM, to produce an optimal solution.

Due to the low change ratio of the synthetic Cili dataset, there is a lot of helpful temporal information in the historical thematic map. The difference in algorithm accuracy reflects the algorithm's ability to utilize the temporal information. From the statistical results reported in Table I, we can find that SLCCM and SPT obtain the lowest OAs among the spatiotemporal SPM methods, indicating that the subpixel change detection and the distance weighted temporal attractiveness are less effective in the utilization of the temporal information. In addition, the proposed SSM_MO, SST_MO, and STSPM_MO methods obtain the best results with a slight accuracy advantage over the latest spatiotemporal SPM methods. SST_MO has a higher OA than that of SSM_MO, which means the temporal prior term plays a more positive role in the multiobjective mapping stage than in the unchanged subpixels detection stage. STSPM_MO has a higher OA than that of SSM_MO and SST_MO, because the temporal information is more fully utilized in two stages than in one stage. Moreover, SSM_MO and SSM_CS, and SST_MO and SST_MAP, respectively, adopted the same spatiotemporal mapping strategies, but the difference lies in SSM_MO and SST_MO using the MOEA while SSM_CS and SST_MAP using the single objective optimization. Therefore, SSM_MO's higher accuracy than SSM_CS and SST_MO's higher accuracy than SST_MAP both suggest that MOEA can better solve the constraint problem of spatiotemporal SPM.

B. Experiments on Real Datasets

Two real datasets composed of multisource remote sensing images were used to examine the validation of SPM

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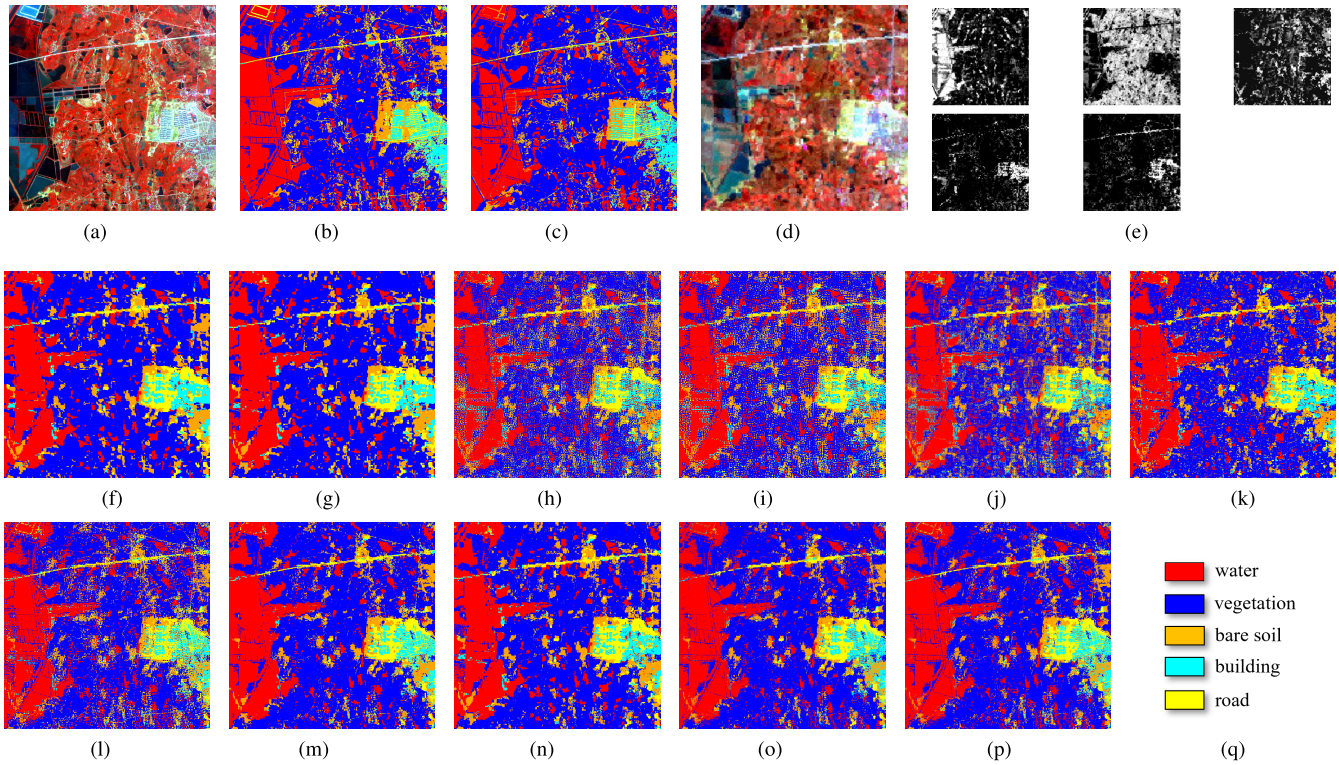


Fig. 9. SPM results for the Wuhan dataset. (a) April 22, 2015, Gaofen-2. (b) April 22, 2015, historical thematic map. (c) April 21, 2016, reference. (d) March 2, 2016, Landsat-8. (e) March 2, 2016, abundance maps of the five component classes. (f) MAP. (g) MOSM. (h) SLCCM. (i) SRM_STD. (j) SSM_CS. (k) SPT. (l) STSRM. (m) SST_MAP. (n) SSM_MO. (o) SST_MO. (p) STSPM_MO. (q) Color legend of five classes.

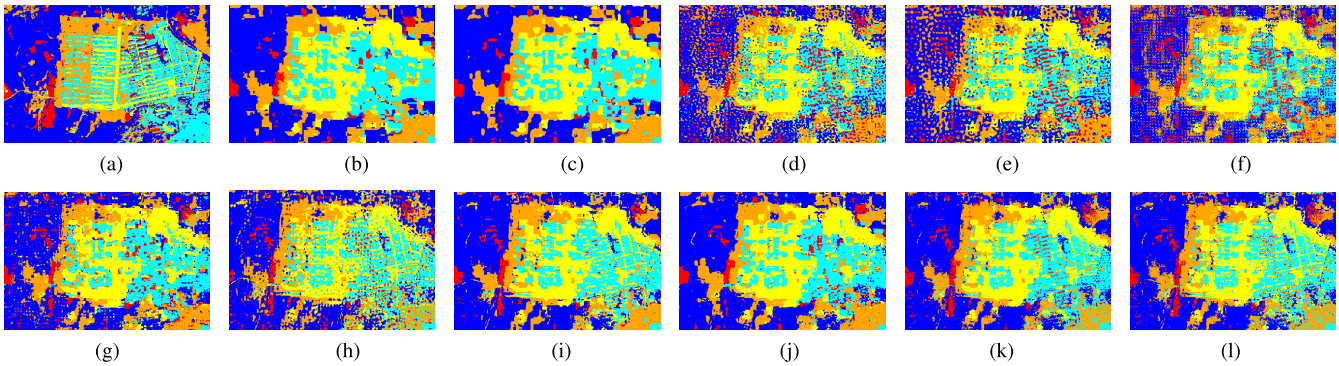


Fig. 10. Zoomed-in areas of the SPM results for the Wuhan dataset. (a) Reference. (b) MAP. (c) MOSM. (d) SLCCM. (e) SRM_STD. (f) SSM_CS. (g) SPT. (h) STSRM. (i) SST_MAP. (j) SSM_MO. (k) SST_MO. (l) STSPM_MO.

methods in practical applications affected by the unmixing errors and the registration errors. In the first real experiment, a multispectral image acquired by the Landsat-8 satellite on March 2, 2016, in Wuhan, was adopted as the coarse image [Fig. 9(d)]. Another multispectral image acquired by the Gaofen-2 satellite on April 22, 2015 was adopted as the historical image [Fig. 9(a)], and its SVM classification result was used as the historical thematic map [Fig. 9(b)]. The study area covers 160 × 160 × 7 pixels in the Landsat-8 image and 960 × 960 × 4 pixels in the Gaofen-2 image, so the mapping scale was set to 6. Registration between the multisource images was conducted in ENVI by manually selecting a group of ground control points with a registration error (rms) less than 0.1. Five classes were manually selected from Landsat-8 image: water, vegetation, bare soil, building, and road. The fully constrained least squares (FCLS) algorithm [39] was then employed to estimate the abundance map of each class [Fig. 9(e)]. For the quantitative assessment of the mapping results, the SVM classification map of another Gaofen-2 image acquired on April 21, 2016, at the same scene, was utilized as the reference map [Fig. 9(c)], whose classification accuracy is 96.79%. The change ratio between the historical thematic map [Fig. 9(b)] and the reference map [Fig. 9(c)] was 32.56%. The change ratios for water, vegetation, bare soil, buildings, and roads are 52.31%, 20.85%, 31.24%, 68.40%, and 83.32%, respectively. Mapping results for the Wuhan dataset are shown in Figs. 9 and 10.

Another real dataset was constructed based on the Landsat-8 image and Sentinel-2A image covering Nanjing City. The generation of Nanjing dataset is as same as the Wuhan dataset. As shown in Fig. 11(a)–(e). The coarse-resolution image obtained by Landsat-8 covering 140 × 140 × 7 pixels.

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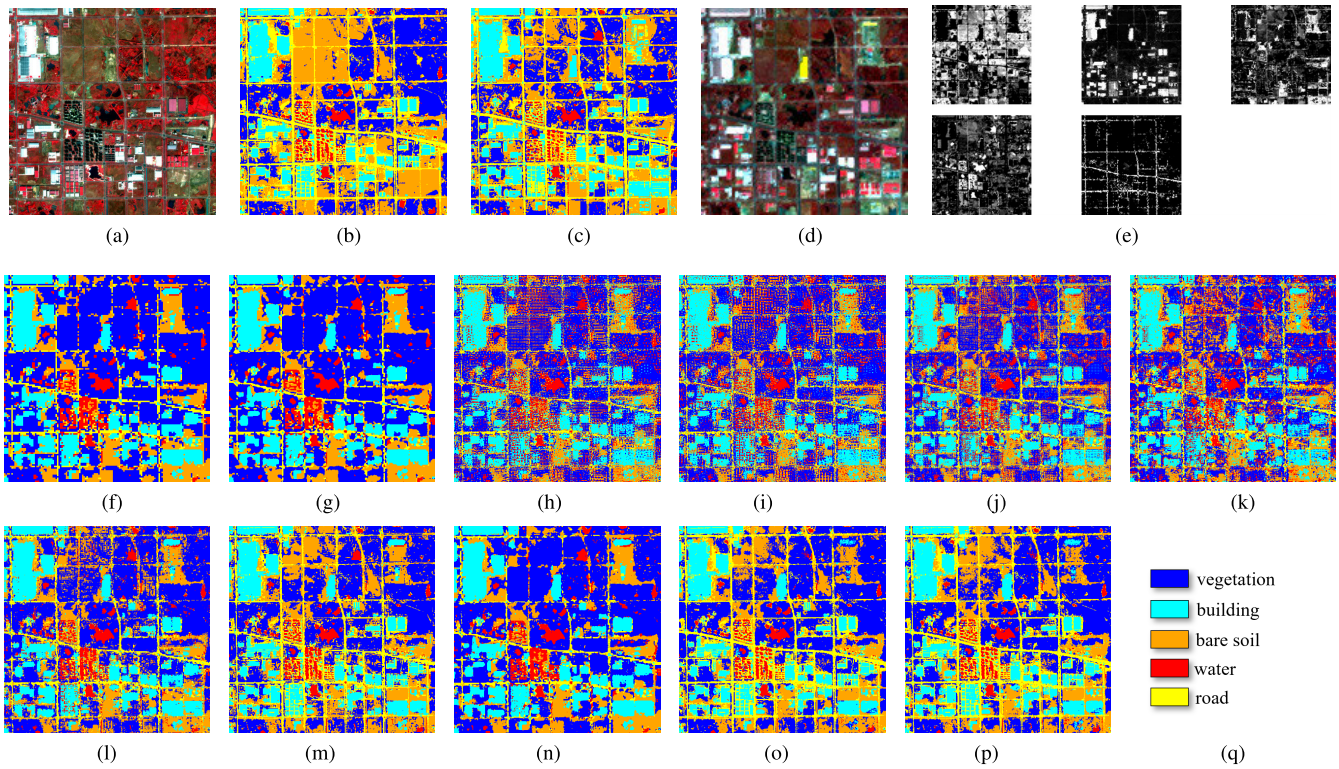


Fig. 11. SPM results for the Nanjing dataset. (a) November 13, 2016, Sentinel-2A. (b) November 13, 2016, historical thematic map. (c) October 9, 2017, reference. (d) October 9, 2017, Landsat-8. (e) October 9, 2017, abundance maps of the five component classes. (f) MAP. (g) MOSM. (h) SLCCM. (i) SRM_STD. (j) SSM_CS. (k) SPT. (l) STSRM. (m) SST_MAP. (n) SSM_MO. (o) SST_MO. (p) STSPM_MO. (q) Color legend of five classes.

The fine-resolution obtained by Sentinel-2A covering $420 \times 420 \times 4$ pixels. The mapping scale was set to 3. The SVM classification accuracy of the reference map [Fig. 11(c)] is 94.46%. Five abundance maps were obtained from the Landsat-8 image by FCLS, including vegetation, buildings, bare soil, water, and roads. During the period from November 13, 2016 to October 9, 2017, lots of changes happened in this area, caused by new buildings and increased vegetation. However, the road network was basically unchanged. The change ratio between the historical thematic map [Fig. 11(b)] and the reference map [Fig. 11(c)] was 36.18%. The change ratios for vegetation, building, bare soil, water, and road are 33.87%, 11.01%, 46.97%, 32.69%, and 8.26%, respectively.

Visual results for the real datasets are provided in Figs. 9–12. In the figures, we can see that large numbers of subpixels are assigned to the wrong classes in the mapping results of SLCCM, SRM_STD, SSM_CS, and SPT. For example, the misclassification of water in the Wuhan dataset, and the false alarms of buildings in the Nanjing dataset. Because these methods rely on the abundance constraint (1) to calculate the number of classes and identify the unchanged subpixels, so they are sensitive to the potential errors of the abundance maps. MAP and MOSM obtain relatively smooth results, thanks to the spatial prior (3). The maps obtained by SST_MAP, SST_MO, and STSPM_MO are more consistent with the reference. They integrate the temporal information into the SPM model, thereby performing better in the preserving the spatial details. However, the result of SSM_MO is worse than those of SST_MO and STSPM_MO, because SSM_MO uses the temporal information only in the unchanged subpixels detection stage. Affected by the inaccurate abundance values, SSM_MO misjudges the unchanged subpixels and retains a lot of erroneous historical information. In addition, the zoomed-in areas of the Wuhan dataset have obvious land-cover changes between buildings and roads (change ratios are 68.4% and 83.32%). The comparison spatiotemporal SPM methods have difficulties in balancing the impact of changed land cover, leading to obvious visual errors in their mapping results. In contrast, the proposed multiobjective optimization-based methods can better balance the influence of the spatial and temporal information, thereby producing more accurate class distribution details of roads and buildings. From the mapping results for the Nanjing dataset, we can see that STSRM, SST_MAP, SST_MO, and STSPM_MO effectively retain the unchanged road network, while accurately classifying the new buildings.

The statistical results of the real datasets are listed in Tables II and III. For the Wuhan dataset, SST_MO and STSPM_MO achieve the optimal and suboptimal results because them consider the temporal information in the multiobjective mapping stage, which is more robust to the input abundance errors and the large land-cover changes. SSM_MO's identification of unchanged subpixels is inaccurate due to the influence of abundance errors. Similarly, for the Nanjing dataset, STSPM_MO and SST_MO achieve the highest OA because their multiobjective mapping stage effectively identified and preserved the road with little change through

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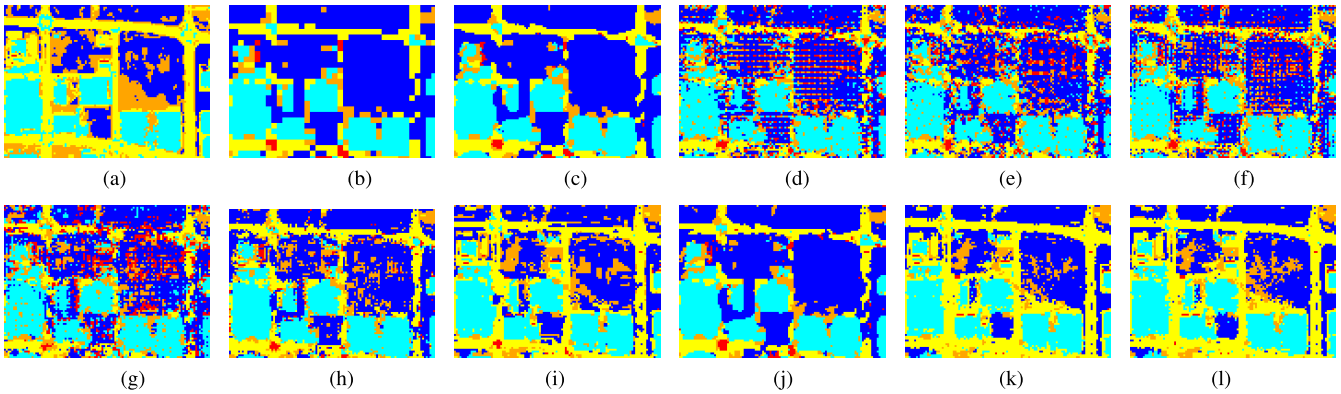


Fig. 12. Zoomed-in areas of the SPM results for the Nanjing dataset. (a) Reference. (b) MAP. (c) MOSM. (d) SLCCM. (e) SRM_STD. (f) SSM_CS. (g) SPT. (h) STSRM. (i) SST_MAP. (j) SSM_MO. (k) SST_MO. (l) STSPM_MO.

TABLE II
MAPPING ACCURACIES FOR THE WUHAN DATASET

	MAP, 2015	MOSM, 2018	SLCCM, 2010	SRM_STD, 2014	SSM_CS, 2016	SPT, 2016	STSRM, 2017	SST_MAP, 2019	**SSM_MO**	**SST_MO**	**STSPM_MO**
Water	56.15	55.89	52.15	53.15	53.31	57.37	61.13	63.58	56.28	66.02	66.15
Vegetation	74.64	74.48	60.86	60.85	60.91	70.90	70.96	76.23	73.67	76.58	75.99
Bare soil	31.59	31.48	28.38	28.84	28.75	28.54	33.43	33.41	31.76	37.32	35.88
Building	30.22	30.17	26.67	26.68	26.69	31.34	30.91	29.55	29.75	28.34	28.83
Road	33.71	33.19	31.04	31.32	31.37	31.60	31.61	33.53	31.16	30.35	31.91
OA (%)	62.55	62.36	53.12	53.42	53.50	60.45	61.78	65.35	61.88	**66.14**	65.87
AA (%)	45.27	45.04	39.82	40.17	40.21	43.95	45.61	47.26	44.52	47.72	**47.75**
Kappa	0.3762	0.3733	0.2717	0.2766	0.2776	0.3538	0.3791	0.4212	0.3679	**0.4329**	0.4315

TABLE III
MAPPING ACCURACIES FOR THE NANJING DATASET

	MAP, 2015	MOSM, 2018	SLCCM, 2010	SRM_STD, 2014	SSM_CS, 2016	SPT, 2016	STSRM, 2017	SST_MAP, 2019	**SSM_MO**	**SST_MO**	**STSPM_MO**
Vegetation	88.30	88.66	72.20	72.88	70.98	67.44	84.09	81.58	88.15	86.23	84.19
Building	74.42	75.54	69.57	69.99	70.22	68.19	77.36	75.92	76.85	73.61	75.35
Bare soil	34.82	34.82	35.93	36.83	36.81	38.05	46.66	54.87	38.26	47.47	49.83
Water	66.34	66.86	62.47	63.92	60.82	58.45	77.25	82.22	71.38	79.71	80.37
Road	19.55	19.03	15.66	17.25	17.13	17.52	31.42	55.83	19.40	67.76	66.64
OA (%)	56.62	57.04	50.18	51.07	50.38	49.28	61.85	67.97	58.13	69.03	**69.19**
AA (%)	56.69	56.98	51.17	52.17	51.19	49.93	63.36	70.08	58.81	70.96	**71.28**
Kappa	0.4139	0.4196	0.3469	0.3570	0.3496	0.3365	0.4883	0.5707	0.4353	0.5862	**0.5886**

the dynamic fusion of multiple constraints. The unchanged subpixels detection stage in SSM_MO failed to retain the road due to the abundance of errors. In addition, for the land cover with high change ratios, such as bare soil, building, road in the Wuhan dataset, and vegetation, building, and water in the Nanjing dataset, the PAs of some spatiotemporal SPM algorithms (SLCCM, SRM_STD, SSM_CS, and SSM_MO) are even lower than the traditional single image SPM methods (MAP and MOSM). This indicates that the temporal information can have a negative effect on SPM when land cover had significantly changed.

C. Parameter Sensitivity Analysis

There are several prior parameters introduced to control the multiobjective optimization process, including the population size NP, the scale factor F, and the crossover rate CR. A unified parameter setting ($NP = 50$, $F = 0.8$, and $\mathrm{CR} = 0.5$) was applied in the above experiments. To demonstrate the stability of the proposed STSPM_MO method, it was necessary to test the sensitivity of these control parameters on both synthetic and real datasets. As for the NP, we fixed the F and CR values and adjusted the NP over the range of 30–150. Mapping accuracies for the Cili and Nanjing datasets are displayed in Fig. 13(a). We found that the OA increases with the expansion of the population when NP is relatively small, and then slightly decreases when NP is larger than 70. However, the variation of the OA is less than 0.8%. We also analyzed the influence of F and CR within reasonable ranges, i.e., $F \in [0.5,\ 1.0]$, $\mathrm{CR} \in [0.1,\ 0.6]$. The quantitative results are displayed in Fig. 13(b) and (c). The low variation (<0.3%) of the proposed method is acceptable.

In addition, the stop criterion in this article is set to maxitr = 30, because the high-dimensional solution is separated into many low-dimensional mapping blocks to be

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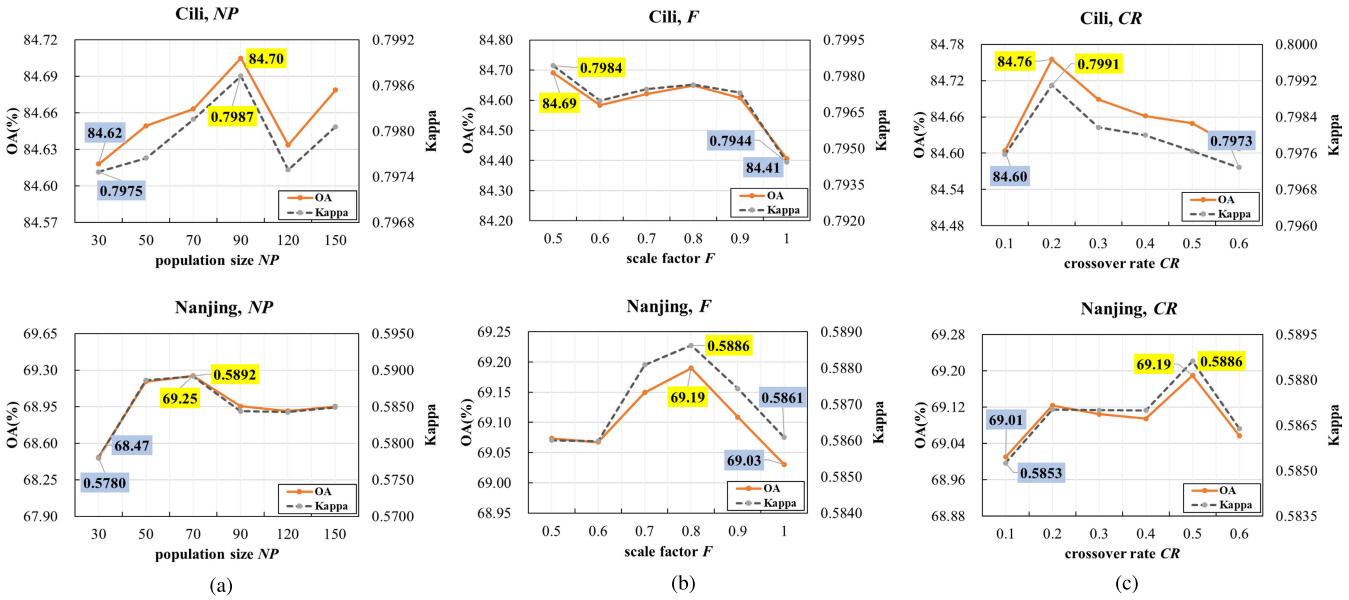


Fig. 13. Parameter sensitivity analysis for STSPM_MO. (a) Sensitivity analysis for NP. (b) Sensitivity analysis for F. (c) Sensitivity analysis for CR.

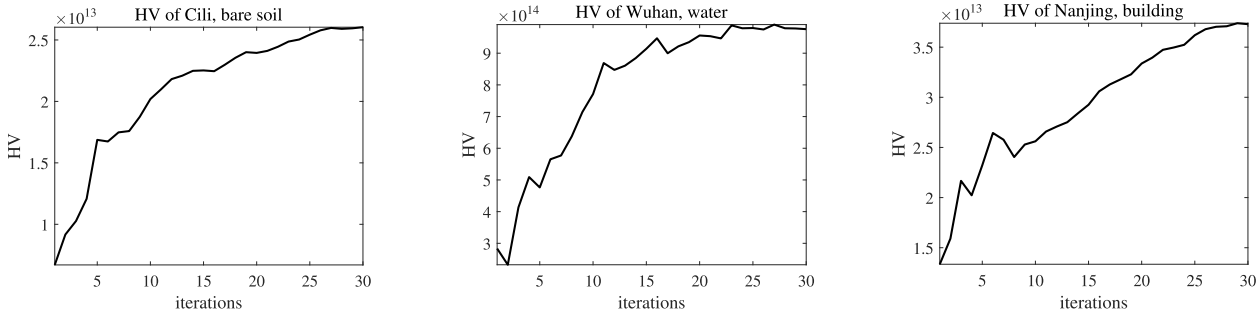


Fig. 14. During the optimization of the class distribution map, the HV of the population gradually increases and then converges after 25 iterations. (a) Cili, bare soil. (b) Wuhan, water. (c) Nanjing, vegetation.

optimized, which can converge in a few iterations. The experimental observation also shows that the population of mapping solutions becomes stable after 25 iterations. To prove the rationality of the stop criterion, hypervolume (HV) [40], [41], which is used to measure the quality of the population convergence, is calculated after each iteration of STSPM_MO. The test result of three datasets is depicted in Fig. 14. We can see that HV increases with iterations, which means that the population has generated better solutions with higher mapping accuracy during evolution. The improvement of the HV gradually stops after 25 iterations, which means that the evolution of the population has converged. For other datasets, using larger iterations as the stopping conditions can ensure the convergence of population, but also increase time cost.

D. Computation Time

To simultaneously optimize a set of high-dimensional SPM solutions, the expensive computational cost can be considered as the "fly in the ointment" of the proposed method. In fact, the computational cost is a common problem in MOEAs and it exponentially increases with the dimension of the solution (image size). To further illustrate this problem, the runtimes of the SPM methods are listed in Table IV. All the experiments were conducted by a personal computer with a 2.3-GHz CPU and 8 GB of RAM, using MATLAB R2015b. The result shows that SLCCM is the most efficient algorithm, owing to its simple rule without model optimization step. STSRM is also efficiency because of its simple optimization step. The processing time of the MAP-based algorithms (MAP and SST_MAP) is acceptable when dealing with small images, but rapidly increases with the image size. Similarly, SPT and SRM_STD, which are both optimized by simulated annealing algorithm, need a long time in the mapping of the Wuhan dataset. SSM_CS is less time consuming, because of its pixelwise optimization mechanism. The MOEA-based methods (MOSM, SSM_MO, SST_MO, and STSPM_MO) are low efficiency, especially for Nanjing and Wuhan datasets. The proposed methods are more time-consuming than MOSM because the clone of multiple solutions significantly increases the computational burden. It will be of great interest to explore more effective and efficient strategies to accelerate the optimization of the multiobjective mapping model. Since the MOEAs are a sort of population-based optimization method,

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TABLE IV
COMPUTATION TIME(S) OF THE SPM METHODS FOR PROCESSING DIFFERENT DATASETS

	MAP, 2015	MOSM, 2018	SLCCM, 2010	SRM_STD, 2014	SSM_CS, 2016	SPT, 2016	STSRM, 2017	SST_MAP, 2019	**STSPM_MO**	**STSPM_MO***
Cili	67.83	319.87	3.21	1034.04	107.35	453.41	22.21	76.03	7561.91	1012.76
Wuhan	37380.87	14327.79	196.92	42610.62	1865.72	24247.80	351.96	39402.54	64557.10	11173.22
Nanjing	2528.90	2181.68	14.29	4238.78	848.72	650.40	69.07	3056.18	39496.18	6566.86

they have implicit parallel processing capability. Therefore, parallel computing can be a promising technique to improve the mapping efficiency. To prove this, the STSPM_MO was speeded up by parallel computing with an eight-core CPU, and its computation time marked as STSPM_MO* was significantly reduced.

E. Discussion

1) Multiobjective Framework for the Spatiotemporal SPM: In the proposed STSPM_MO method, the abundance constraint, spatial dependence constraint, and temporal dependence constraint of the spatiotemporal SPM problem are modeled as three objective functions, whose information is automatically fused by the MOEA. The main idea of our work is to use multiobjective optimization to appropriately describe and solve the constrained problem of spatiotemporal SPM. STSPM_MO is a novel framework for spatiotemporal SPM, in which the spatial and temporal prior terms can be replaced by other suitable models, such as spatial attraction [42], [43], the total variation-based spatial prior [10] or the temporal dependence index [15]. The MOEA can also be replaced by other multiobjective optimization algorithms [44], [45].

2) Functions of the Three Constraints in STSPM_MO: From the above experiments, we can conclude that the abundance, spatial, and temporal information play different roles when dealing with different datasets. To be more specific, the relaxed abundance constraint is robust to abundance maps with potential errors. The spatial information tends to generate a spatially smooth result. The temporal information is extremely important for the reconstruction of subpixel distribution details. In STSPM_MO, the temporal information is fully utilized in both the unchanged subpixels detection step and the multiobjective mapping step. We found that the temporal information for each step makes different contributions to the mapping results. For instance, in the synthetic experiment with accurate abundance maps as input, the unchanged subpixels detection step can effectively preserve the fine spatial distribution details, so SSM_MO obtains a similar OA to that of SST_MO and STSPM_MO. However, when the abundance maps are inaccurate in the two real experiments, the unchanged subpixels detection step will be invalid, while the temporal prior term in the multiobjective mapping step can alternatively use the valid historical land-cover information and is less sensitive to the abundance errors, so SST_MO and STSPM_MO obtain much higher OAs than SSM_MO.

3) Influence of the Land-Cover Change Ratio: Except for the abundance errors, the land-cover change ratio is another important factor for the quantity of temporal information. In general, changes between the bitemporal images reduce the effectiveness of the temporal information [46], so the mapping accuracy increases as the total change ratio between the bitemporal images decreases. For example, all the tested spatiotemporal SPM methods perform better than the traditional single image SPM methods (MAP and MOSM) in the experiment of the Cili dataset, while some of them (SLCCM, SRM_STD, SSM_CS, and SPT) obtain lower OAs than traditional methods in the two real experiments. One of the important reasons is that the total change ratio of the Cili dataset (20.54%) is smaller than that of the Wuhan and Nanjing datasets (32.56% and 36.18%). Furthermore, the change ratio of each class also has an impact on its PA improvement, but it is not proportional. For example, the rate of building and road cover changes in the Wuhan dataset is too high (68.40% and 83.32%), which leads to a decrease in PAs of some spatiotemporal SPM algorithms (SLCCM, SRM_STD, SSM_CS, SST_MAP, SSM_MO, SST_MO, and STSPM_MO) compared with traditional methods. The change ratio of vegetation, urban, and bare soil in the Cili dataset is relatively small (13.89%, 9.35%, and 15.30%), so the spatiotemporal SPM algorithms have PAs improvement on these classes. Water in the Cili dataset has the smallest change ratio (4.94%), but its PA improvement is not the most significant. In other words, the temporal information can help allocate the changed pixels when the land-cover change ratio is small, or conversely, may complicate the allocation of changed pixels when land-cover changes are extensive. The key is whether the spatiotemporal SPM methods can accurately identify the unchanged subpixels and effectively utilize the temporal information.

4) Visual Judgment in STSPM_MO: Visual judgment is used to select the best distribution map from the final population. In the final population, there are NP mapping solutions that achieve different balances of the abundance, spatial, and temporal constraints. Correspondingly, these solutions have different objective function values in the multiobjective optimization model (7). Visually, each solution is different in the spatial smoothness and the detail fidelity. We choose the ones that are spatially smooth in the homogeneous areas and spatially rich in detail at the edges as the optimal solution, i.e., solutions with the three objective function values closer to the zero point (0, 0, 0). Actually, after multiple iterations of optimization, the objective function values of the solutions in the final converged population are mainly distributed close to the zero point. By visually selecting solutions, the solutions with the smallest single objective function value can be avoided, i.e., solutions with over-smooth spatial distribution or

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spatial distribution too similar to the historical thematic map. Therefore, selecting different solutions has a relatively small impact on the mapping accuracy, as the resulting subpixel classification maps are similar.

5) Potential Improvement Strategies: The mapping accuracy improvement or deterioration of the spatiotemporal SPM algorithms both affected by the abundance errors and the land-cover change ratio, so it will be a meaningful task to prejudge the characteristics of the input data and then automatically adjust the effects of the abundance and spatiotemporal information. For example, by prejudging the abundance errors using the root-mean-square error of the spectral unmixing and the geometric registration error of the bitemporal images, then the impact of abundance constraints on SPM can be adaptively adjusted based on the abundance errors. Alternatively, by predicting the land-cover change ratio using the time interval between the bitemporal images and the land-cover types, then the impact of temporal constraint on SPM can be adaptively adjusted based on the change ratio. Our work attempts to dynamically integrate different information using multiobjective optimization theory. However, we have not considered the characteristics of the data, which will be addressed in the future. In addition, due to the complicated imaging environment, the remote sensing images in the real datasets contain both linear and nonlinear spectral mixed pixels. However, our work only considered the linear mixture in the modeling of spectral unmixing and SPM. There are two ways to take the nonlinear mixed problem into account. One way is to use a nonlinear spectral unmixing model to obtain the abundance maps of the nonlinear mixed classes as the input data for SPM, such as the generalized bilinear model [47] and the artificial neural networks [48]. The other way is to modify the spatial constraint to enhance the nonlinear fitting ability of the SPM model, such as the point spread function [49], [50], the radial basis functions [51], and the learnable spatial correlation based on the convolutional neural networks [52]. However, the proposed method, as well as most of the SPM methods uses the spatial correlation to construct the spatial constraint, which is designed for the linear mixed classes. Developing a spatial constraint for the nonlinear mixed classes would be a potential way to improve the mapping accuracy of the proposed method.

V. Conclusion

In this article, we have proposed a novel spatiotemporal SPM method based on multiobjective optimization theory (STSPM_MO), in which the constrained optimization problem of spatiotemporal SPM is converted into an MOP with three objective functions. The MOEA is employed to simultaneously optimize the multiple objectives and fuse their information to produce the optimal class distribution maps. Compared with the latest spatiotemporal SPM methods, STSPM_MO does not use sensitive weights. In addition, in STSPM_MO, the temporal information is fully exploited by the spatiotemporal map fusion and the multiobjective SPM model, and their effectiveness was verified by the two variants of SSM_MO and SST_MO. A synthetic dataset and two real datasets were tested in the experiments. The results proved that the temporal information is beneficial for the construction of spatial distribution details. The superiority of the multiobjective optimization in SPM was demonstrated by the accuracy advantage of SSM_MO over SSM_CS and SST_MO, STSPM_MO over SST_MAP. Furthermore, the experiments found that the proposed methods performed better in eliminating the potential errors of the abundance maps. The stability of STSPM_MO was also confirmed by the sensitivity analysis.

However, the excessive computational complexity hinders the application of the proposed approach to the mapping of large-scale imagery. The best class distribution maps were visually selected from the final solutions, so it will be necessary to develop an adaptive selection strategy by analyzing the characteristics of the input data, to further improve the automation of the proposed approach. In addition, with the rapid increase in the amount of remote sensing images, the spatial distribution patterns are complex and diverse, the presented unsupervised SPM methods, which adopt prior models, have limited accuracy improvement. The deep learning-based remote sensing image processing methods [53], [54], [55], [56], [57], which have powerful feature learning capability, are promising to be applied to spatiotemporal SPM to cope with the massive remote sensing data.

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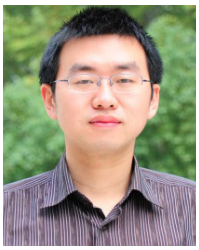
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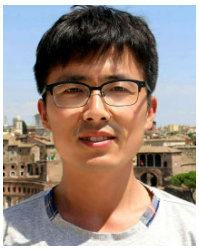
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